

Two errors, not one: the blade tones are bent by up to 8 dB below 1 kHz, and under them sits a broadband error of ± 1.3 dB (2.7 dB worst) that no band is exempt from. **The arc is not qualified to a standard anywhere yet.** For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7-18, 1-2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

remedies report: docs/arc-validation-remedies.pdf



Fig. 2. One fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11-14 mirrored. Rows aligned: mic row i sat at spot row i. 0° mic never moved, so its M and the 0° P are known only as a sum. Third-octave centre, Hz.

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4;
prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.